

ComSci 342 Homework 6

1. [α -renaming] Write an α -equivalent expression for $(((\lambda x. \lambda x. \lambda x. (x \lambda x. \lambda y. (x y))))))$.

$(\lambda a. \lambda b. \lambda c. (c \lambda d. \lambda e. (d e)))$

2. [β -reduction] Perform β -reduction for the following λ expressions.

a. $(((\lambda x. (x x)) (\lambda y. y)) x)$

$$(((\lambda x. (x x)) (\lambda y. y)) x)$$

$$= (((\lambda y. y) (\lambda y. y)) x)$$

$$= ((\lambda y. y) x)$$

$$= x$$

b. $(((\lambda x. (\lambda y. (y y))) (\lambda w. v a)) b)$

$$(((\lambda x. (\lambda y. (y y))) (\lambda w. (v a))) b)$$

$$= ((\lambda y. (y y)) b)$$

$$= (b b)$$

c. $(\lambda x. m (\lambda y. y ((\lambda v. v a) (\lambda w. w b))))$

$$((\lambda v. v a) (\lambda w. w b))$$

$$= ((\lambda w. w b) a)$$

$$= (a b)$$

$$\lambda x. (m (\lambda y. (y (a b))))$$

d. $((\lambda x. (\lambda f. \lambda x. (f (f (f x))) g) z) \lambda f. \lambda x. (f (f x)))$

$$((\lambda x. (((\lambda f. \lambda x. (f (f (f x))) g) z)) (\lambda f. \lambda x. (f (f x))))$$

$$= (((\lambda f. \lambda x. (f (f (f x))) g) z)$$

$$= ((\lambda x. (g (g (g x)))) z)$$

$$= (g (g (g z)))$$

e. $((\lambda m. \lambda n. (x y) ((\lambda a. \lambda b. (a b) \lambda x. x) g)) \lambda z. ((z p) q))$

$((\lambda m. \lambda n. ((x y) (((\lambda a. \lambda b. (a b)) (\lambda x. x)) g))) (\lambda z. ((z p) q)))$

$= \lambda n. ((x y) (((\lambda a. \lambda b. (a b)) (\lambda x. x)) g))$

$((\lambda a. \lambda b. (a b)) (\lambda x. x))$

$= (\lambda b. ((\lambda x. x) b))$

$= (\lambda b. b)$

$\lambda n. ((x y) ((\lambda b. b) g))$

$= (\lambda n. ((x y) g))$

3. [Evaluation order] The goal of this problem is to help you understand the evaluation order of lambda calculus. Show the steps of β -reduction for the following lambda expression using two different evaluation orders.

$(\lambda x. p (\lambda y. (y y) (\lambda z. (z z))))$

Normal Order:

$(\lambda x. p)((\lambda y. (y y))(\lambda z. (z z)))$

$= p$

Applicative Order:

$(\lambda x. p)((\lambda y. (y y))(\lambda z. (z z)))$

$(\lambda y. (y y))(\lambda z. (z z))$

$= (\lambda z. (z z))(\lambda z. (z z))$

$= (\lambda z. (z z))(\lambda z. (z z))$

Normal order: p

Applicative order: diverges / infinite β -reduction

4. [Church Encoding] Encode the logic Boolean operation $a \rightarrow b$ using true, false, ite.

If a is true, result is b. Else if a is false, result is true:

$\lambda a. \lambda b. (((\text{ite } a) b) \text{ true})$

5. [Understanding lambda expressions and Church Encoding] Given:

unknown: $(\lambda n. ((n (\lambda x. \text{true})) \text{ false}))$

a. What is the result of (unknown zero)?

(unknown zero)
= (zero $(\lambda x. \text{true})$ false)
= $((\lambda f. \lambda x. x) (\lambda x. \text{true}) \text{ false})$
= $((\lambda x. x) \text{ false})$
= false

b. What is the result of (unknown one)?

(unknown one)
= (one $(\lambda x. \text{true})$ false)
= $((\lambda f. \lambda x. (f x)) (\lambda x. \text{true}) \text{ false})$
= $((\lambda x. ((\lambda x. \text{true}) x)) \text{ false})$
= $((\lambda x. \text{true}) \text{ false})$
= true

c. What is the result of (unknown two)?

(unknown two)
= (two $(\lambda x. \text{true})$ false)
= $((\lambda x. \text{true}) ((\lambda x. \text{true}) \text{ false}))$
= true

d. What computation does unknown perform?

It returns false for zero and true for any other Church numeral.

- e. Using unknown to define a function IsOpposite. It tests if two numbers m and n are opposite (two numbers are opposites if they have the same absolute value but different signs. For example, 4 and -4, and 1 and -1 are opposite).

pair := $\lambda x.\lambda y.\lambda z.((z\ x)\ y)$

fst := $\lambda p.(p\ \text{true})$

snd := $\lambda p.(p\ \text{false})$

eqNat := $\lambda m.\lambda n.((\text{and}\ (\text{leq}\ m\ n))\ (\text{leq}\ n\ m))$

leq := $\lambda m.\lambda n.(\text{isZero}\ (\text{sub}\ n\ m))$

isZero := $\lambda n.((n\ (\lambda x.\text{false}))\ \text{true})$

sub := $\lambda m.\lambda n.((n\ \text{pred})\ m)$

pred := $\lambda n.\lambda f.\lambda x.(n\ (\lambda g.\lambda h.(h\ (g\ f)))\ (\lambda u.x)\ (\lambda u.u))$

sign := $\lambda n.(\text{unknown}\ (\text{fst}\ n))$

abs := $\lambda n.(((\text{sign}\ n)\ n)\ (\text{pair}\ (\text{snd}\ n)\ (\text{fst}\ n)))$

eq := $\lambda m.\lambda n.\text{and}\ (\text{eqNat}\ ((\text{fst}\ m)\ (\text{snd}\ n))((\text{fst}\ n)\ (\text{snd}\ m)))$

IsOpposite :=

$\lambda m.\lambda n.(\text{and}\ (\text{eq}\ (\text{abs}\ m)\ (\text{abs}\ n))(\text{and}\ (\text{unknown}\ (\text{fst}\ (\text{abs}\ m)))(\text{not}\ (\text{eqNat}\ (\text{fst}\ m)\ (\text{fst}\ n))))$

6. [Church Encoding, understanding the semantics of lambda expressions] Given:

data: $(\lambda x.(\lambda y.(\lambda z.((z\ x)\ y))))$

op1: $(\lambda p.(p\ (\lambda x.(\lambda y.x))))$

op2: $(\lambda p.(p\ (\lambda x.(\lambda y.y))))$

true: $(\lambda x.(\lambda y.x))$

false: $(\lambda x.(\lambda y.y))$

a. What is the result of (op1 ((data a) b))?

$((\text{data } a) b)$
 $= ((\lambda x.\lambda y.\lambda z.((z\ x)\ y))\ a\ b)$
 $= \lambda z.((z\ a)\ b)$
 $(\text{op1 } ((\text{data } a) b))$
 $= (\lambda p.(p\ (\lambda x.\lambda y.x)))\ (\lambda z.((z\ a)\ b))$
 $= ((\lambda z.((z\ a)\ b))\ (\lambda x.\lambda y.x))$
 $= (((\lambda x.\lambda y.x)\ a)\ b)$
 $= ((\lambda y.a)\ b)$
 $= a$

b. What is the result of (op2 ((data a) b))?

$(\text{op2 } ((\text{data } a) b))$
 $= (\lambda p.(p\ (\lambda x.\lambda y.y)))\ (\lambda z.((z\ a)\ b))$
 $= ((\lambda z.((z\ a)\ b))\ (\lambda x.\lambda y.y))$
 $= (((\lambda x.\lambda y.y)\ a)\ b)$
 $= ((\lambda y.y)\ b)$
 $= b$

c. What computation do op1 and op2 perform?

op1 returns the first element of the pair whereas, op2 returns the second element of the pair.